SQL SYLLABUS FOR APPLICATION/PRODUCTION SUPPORT ENGINEER

Basics of SQL

- Introduction to RDBMS (Relational Database Management System)
- SQL vs NoSQL databases
- Common databases: MySQL, PostgreSQL, MariaDB, Oracle, SQL Server

SQL Queries & Operations

- Basic Queries: SELECT, FROM, WHERE, ORDER BY, GROUP BY, HAVING
- Joins: INNER JOIN, LEFT JOIN, RIGHT JOIN, FULL JOIN, SELF JOIN
- Subqueries & Nested Queries
- Data Manipulation Language (DML):
    - INSERT, UPDATE, DELETE
- Data Definition Language (DDL):
    - CREATE TABLE, ALTER TABLE, DROP TABLE
- Data Control Language (DCL):
    - GRANT, REVOKE
- Transaction Control:
    - COMMIT, ROLLBACK, SAVEPOINT

Indexes & Performance Optimization

- Indexing (CREATE INDEX, DROP INDEX)
- Query Optimization (EXPLAIN, ANALYZE)
- Caching mechanisms

Stored Procedures & Functions

- Writing and executing stored procedures
- Triggers and Events

Backup & Recovery

- Database backups: mysqldump, pg_dump
- Restoring databases

User Management & Security

- Creating and managing users (CREATE USER, ALTER USER)
- Assigning roles and privileges
- Auditing and logging activities

Common SQL Issues & Troubleshooting

- Deadlocks and locking issues
- Debugging slow queries
- Connection pool management

LINUX SYLLABUS FOR APPLICATION/PRODUCTION SUPPORT ENGINEER

Linux Fundamentals

- Linux distributions (Ubuntu, CentOS, RHEL, Debian)
- Linux architecture (Kernel, Shell, File System)
- Boot Process and System Initialization

File System & Permissions

- File system hierarchy
- File and directory operations (ls, cp, mv, rm)
- File permissions and ownership (chmod, chown, chgrp)
- Hard and soft links (ln)

Process Management

- Listing and monitoring processes (ps, top, htop)
- Killing or stopping processes (kill, pkill, killall)
- Background and foreground processes (&, nohup, jobs, fg, bg)

User & Group Management

- Creating and managing users (useradd, usermod, passwd, deluser)
- Managing groups (groupadd, groupmod, groups)
- Setting permissions for users and groups

Networking & Connectivity

- Network configuration (ifconfig, ip, netstat, ss)
- Checking connectivity (ping, traceroute, telnet, curl, wget)
- Managing firewall (iptables, firewalld, ufw)
- SSH (Secure Shell) configuration and usage (ssh, scp, sftp)

Log Management & Monitoring

- Log files (/var/log/, journalctl, syslog)
- Monitoring system performance (vmstat, iostat, sar)
- Disk usage (df, du)

Shell Scripting & Automation

- Writing basic shell scripts (bash, sh)
- Using loops and conditional statements
- Scheduling tasks with cron and at
- Automating log cleanup, backups, and monitoring

Package Management

- Using package managers:
  - apt, dpkg (Debian/Ubuntu)
  - yum, rpm, dnf (RHEL/CentOS)
- Installing and removing software

Database & Server Management

- Connecting Linux to a database (MySQL, PostgreSQL)
- Running and managing database services (systemctl, service)
- Managing web servers (Apache, Nginx)

Backup & Recovery

- File and system backup (tar, rsync, scp)
- Disaster recovery techniques

Common Linux Issues & Troubleshooting

- System resource bottlenecks (CPU, RAM, Disk I/O)
- Debugging application crashes
- Analyzing logs for failures
- Resolving user permission issues

SQL STUDY RESOURCES & PRACTICE TASKS

Study Resources

- Books:
  - SQL for Beginners – Jon Duckett
  - SQL in 10 Minutes, Sams Teach Yourself – Ben Forta
  - High Performance MySQL – Baron Schwartz
- Video Courses:
  - SQL for Beginners (freeCodeCamp)
  - The Complete SQL Bootcamp (Udemy)
- Interactive Platforms:
  - SQLZoo
  - LeetCode SQL Problems
  - Mode Analytics SQL Tutorial

Practice Tasks

- Basic Queries
  - Retrieve all records from a table using SELECT *.
  - Filter data using WHERE (e.g., employees with salary > 50000).
  - Sort data using ORDER BY (e.g., by name or date).
- Joins & Subqueries
  - Get a list of employees along with their department names (using JOIN).
  - Find employees earning above the company's average salary (using SUBQUERY).
- Performance Optimization
  - Create an index on a column and measure the difference in query speed.
  - Use EXPLAIN to analyze query execution plans.
- Stored Procedures & Triggers
  - Write a stored procedure to insert a new user and check if the email exists.
  - Create a trigger that updates a log table whenever a row is deleted.
- Backup & Recovery
  - Perform a full database backup using mysqldump.
  - Restore a database from a backup file.

LINUX STUDY RESOURCES & PRACTICE TASKS

Study Resources

- Books:
  - Linux Command Line and Shell Scripting Bible – Richard Blum
  - The Linux Programming Interface – Michael Kerrisk
- Video Courses:
  - Linux for Beginners (freeCodeCamp)
  - Red Hat Linux System Administration (Udemy)
- Interactive Platforms:
  - OverTheWire: Bandit (Linux Challenges)
  - Linux Survival (Interactive Tutorials)
  - Practice Linux commands in Browser

Practice Tasks

- Basic Linux Commands
  - List all files in a directory (ls -la).
  - Create, move, rename, and delete files (touch, mv, rm).
- File Permissions & User Management
  - Change file permissions (chmod 755 filename).
  - Create a new user and add them to a group (useradd, usermod).
- Process & Performance Monitoring
  - Monitor system resource usage (top, htop, vmstat).
  - Kill a process by PID (kill -9 <pid>).
- Networking
  - Check open ports on the system (netstat -tulnp).
  - Test connectivity to a remote server (ping google.com).
- Shell Scripting & Automation
  - Write a shell script to check disk space and send an alert if low.
  - Automate a database backup using a cron job.
- Log Analysis & Troubleshooting
  - Find the top 10 most common errors in /var/log/syslog.
  - Search for failed login attempts (grep "Failed password" /var/log/auth.log).
- Final Tips
  - Set up a Virtual Machine (VM) or use a Cloud Server to practice Linux (try Ubuntu or CentOS).
  - Use a local database setup (MySQL, PostgreSQL) to experiment with SQL queries.
  - Apply real-world scenarios, like troubleshooting a slow query or debugging a Linux service.